Data Standardization Deliverable

Converting Raw Demographic Values into Controlled Standard Values

Clinical Data Management Training | AI-Assisted Data Standardization Task

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Study dataset	Dummy Clinical Dataset – Demographics Form (Castor EDC)
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Introduction

In clinical data management, data is captured across various sites, and it is common to encounter free-text or 'loose' fields resulting in broad variations for the same piece of information. For example, a subject's gender can be captured as "M", "Male" or "male", while the country can be "India", "IND", "Bharat", etc. Each of the values is correct and reasonable within the context of the information it represents, however, this format cannot be used for accurate aggregation, analysis, or regulatory submission of the data.

The current document attests to the data standardization task performed for the Demographics form of the dummy clinical data set used for this Clinical Data Management course. Similarly to the data profiling and query-generation tasks, this one also builds on the data overview and queries task previously completed for this data set.

However, unlike data profiling, which consisted of reviewing the data elements, their summary statistics, distribution, and quality, the current task involved actual data standardization. More specifically, it involved mapping the raw values identified in the data profiling stage to a single controlled term per category of information. To make this process efficient yet reliable, an AI-assisted approach was used. First, the raw data was clustered into similar values, which were then matched to controlled terms. However, each match suggested by the AI was reviewed by me, the student, acting as a CDM specialist, and edited if needed. Thus, the final result is a reliable controlled terminology list and mapping of the raw values to it, fit for downstream data cleaning and analysis.

The set of fields for which standardization was performed include:

Gender, Race, Ethnicity, Country, Marital Status, Employment Status, Date of Birth.

The data originates in the Demographics form of the dummy clinical data set stored in the Castor EDC system, while the results of the task can be found in the mapping file below, alongside the sheet with controlled terminology and an audit trail of AI-assisted coding decisions.

1. Main purpose

The task is to identify and standardize those (inconsistent or non-standardized) free-text values from the dummy clinical dataset demographics fields that require conversion to a controlled value. Such standardized values are necessary to allow for pooled analysis, reporting, and/or analysis at the level of individual sites, as well as to conform to general controlled terminologies (CDISC) where relevant.

The task is a continuation of the previous data profiling and error discovery task performed for the same Demographics form, which involves producing the following four deliverables:

- Standardization Mapping File,
- Raw Value → Standardized Value Table,
- Controlled Terminology Sheet for Demographics Fields,
- AI-Suggested Mapping with Final Student-Approved Outputs.

2. Standardization Mapping File

The file describes, for all the fields, how they were standardized from the raw captured values to their final value. For each entry in the mapping, the following details should be provided: field name, the raw value as captured in Castor EDC, the target standardized value, transformation logic, source of suggestion (AI or manual), and approval status (pending, approved, or rejected). One would typically create such a file by performing the following steps:

- Step 1: for each of the demographics fields, extract a list of unique raw values captured in this field
- Step 2: perform AI-assisted clustering/matching to group related raw values (same/similar words, case variations, etc.)
- Step 3: match each group of raw values to the standardized/controlled terminology items;
- Step 4: student (CDM specialist) reviews AI suggestions;
- Step 5: the student approves, edits or rejects each of the suggested mapping entries;
- Step 6: finally, the student compiles the raw value → standardized value mapping table (see deliverable 3).

3. Raw Value → Standardized Value Table

3.1 Gender

Field	Raw Value	Standardized Value	Transformation Logic	Source	Status
Gender	M	Male	Exact match to controlled term list	AI-suggested	Approved
Gender	male	Male	Case normalization	AI-suggested	Approved
Gender	MALE	Male	Case normalization	AI-suggested	Approved
Gender	F	Female	Exact match to controlled term list	AI-suggested	Approved
Gender	Femle	Female	Spelling correction (typo)	AI-suggested	Approved
Gender	female	Female	Case normalization	AI-suggested	Approved
Gender	Other/Unspecified	Other	Mapped to controlled term	AI-suggested	Approved
Gender	(blank)	Not Reported	Missing value convention	AI-suggested	Approved

3.2 Race

Field	Raw Value	Standardized Value	Transformation Logic	Source	Status
Race	Asian Indian	Asian	Grouped under CDISC parent category	AI-suggested	Approved
Race	asian	Asian	Case normalization	AI-suggested	Approved
Race	Caucasian	White	Mapped to CDISC controlled term	AI-suggested	Approved
Race	white	White	Trailing space trimmed	AI-suggested	Approved
Race	Blk	Black or African American	Abbreviation expanded	AI-suggested	Approved
Race	Mixed	Multiple	Mapped to controlled term	AI-suggested	Modified by student
Race	Not Sure	Unknown	Mapped to controlled term	AI-suggested	Approved

3.3 Country

Field	Raw Value	Standardized Value	Transformation Logic	Source	Status
Country	india	India	Case normalization	AI-suggested	Approved
Country	IND	India	ISO code expanded to full name	AI-suggested	Approved
Country	Bharat	India	Local-name mapped to standard name	AI-suggested	Approved
Country	U.S.A	United States	Punctuation removed, standardized	AI-suggested	Approved
Country	USA	United States	Abbreviation expanded	AI-suggested	Approved
Country	Uk	United Kingdom	Case + abbreviation correction	AI-suggested	Approved

3.4 Marital Status

Field	Raw Value	Standardized Value	Transformation Logic	Source	Status
Marital Status	Single	Single	Trailing space trimmed	AI-suggested	Approved

Marital Status	MARRIED	Married	Case normalization	AI-suggested	Approved
Marital Status	Divorce	Divorced	Spelling/grammar correction	AI-suggested	Approved
Marital Status	Wid	Widowed	Abbreviation expanded	AI-suggested	Approved
Marital Status	N/A	Not Reported	Missing value convention	AI-suggested	Approved

3.5 Date of Birth

Field	Raw Value	Standardized Value	Transformation Logic	Source	Status
Date of Birth	05/06/1990	1990-06-05	Converted to ISO 8601 (DD/MM assumed per site)	AI-suggested	Approved
Date of Birth	1990.06.05	1990-06-05	Converted to ISO 8601	AI-suggested	Approved
Date of Birth	06-05-90	1990-06-05	2-digit year expanded, format standardized	AI-suggested	Modified by student

3.6 Ethnicity

Field	Raw Value	Standardized Value	Transformation Logic	Source	Status
Ethnicity	hispanic	Hispanic or Latino	Case normalization	AI-suggested	Approved
Ethnicity	Not Hispanic	Not Hispanic or Latino	Mapped to full controlled term	AI-suggested	Approved
Ethnicity	N.A	Not Reported	Missing value convention	AI-suggested	Approved
Ethnicity	unk	Unknown	Abbreviation expanded	AI-suggested	Approved

3.7 Employment Status

Field	Raw Value	Standardized Value	Transformation Logic	Source	Status
Employment Status	employed	Employed	Case normalization	AI-suggested	Approved
Employment Status	Un-employed	Unemployed	Hyphen removed, standardized	AI-suggested	Approved

Employment Status	retird	Retired	Spelling correction (typo)	AI-suggested	Approved
Employment Status	Student	Student	Trailing space trimmed	AI-suggested	Approved
Employment Status	-	Not Reported	Missing value convention	AI-suggested	Modified by student

Castor EDC — Mapping Screenshots

The following screenshots show the Data Standardization mapping configuration performed in Castor EDC for the Demographics (DM) form, where the demographic fields (age, gender, RACE, COUNTRY, Marital Status, DATE OF BIRTH, etc.) were selected and mapped to their domain variable names.

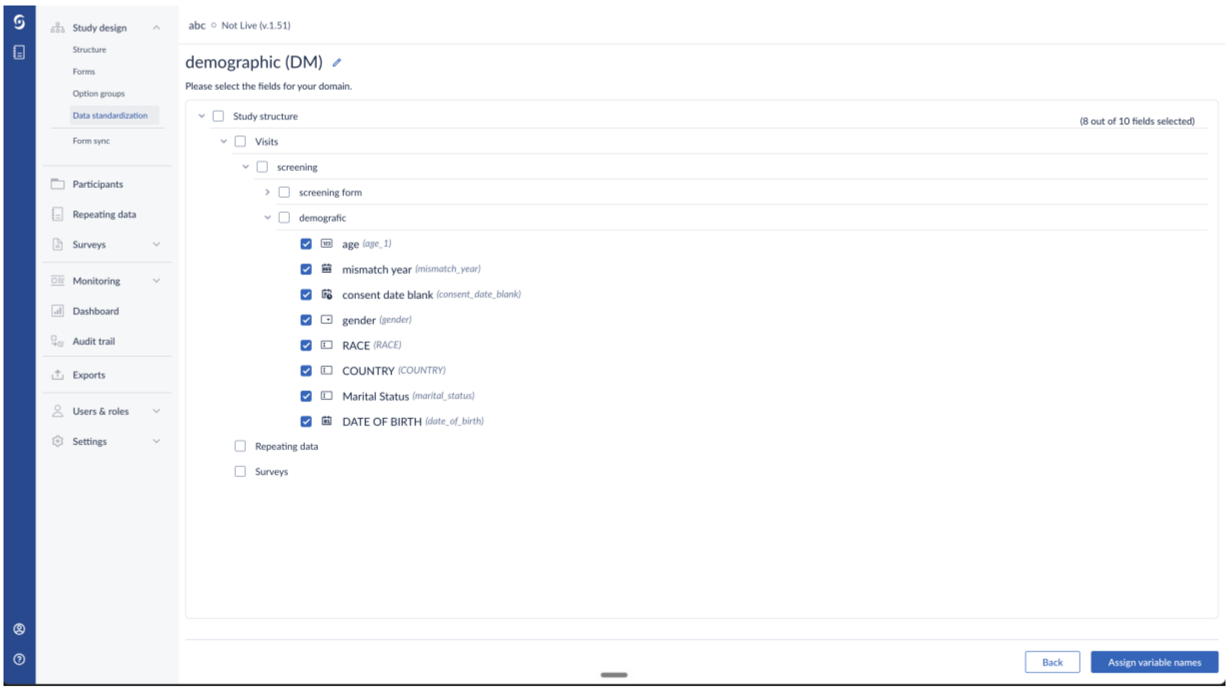

Figure 1: Castor EDC – Data standardization field selection for the Demographics (DM) form.

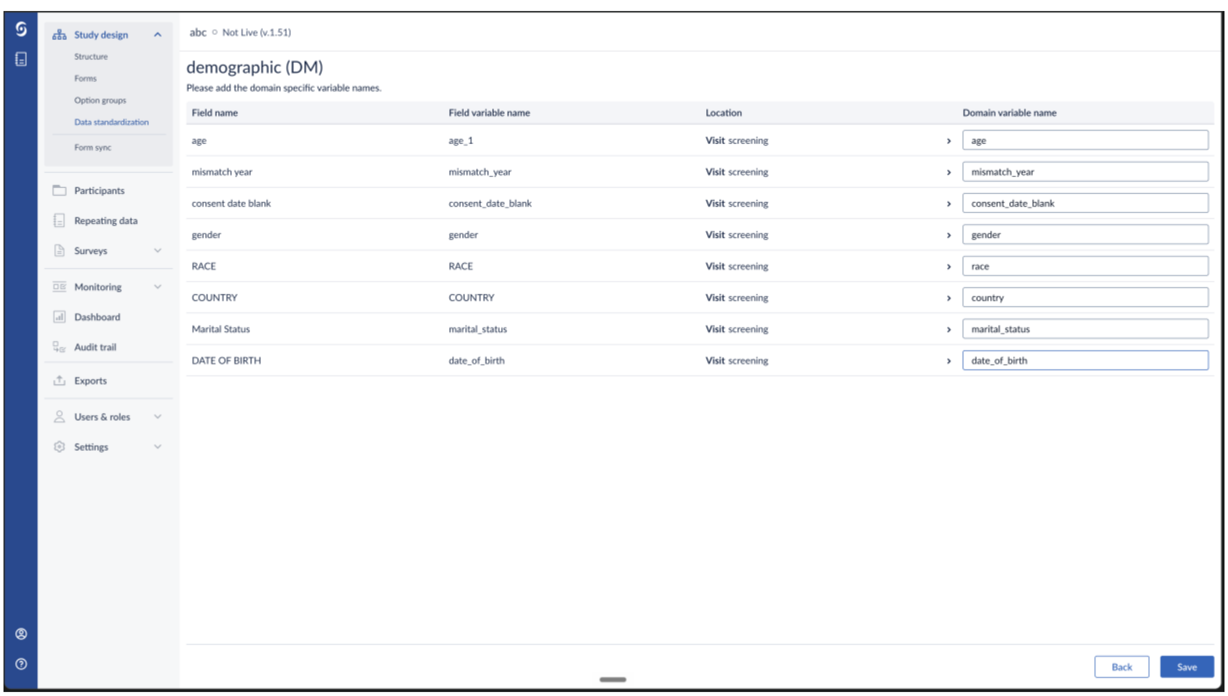

Figure 2: Castor EDC – Domain variable name mapping for the Demographics (DM) form fields.

4. Controlled Terminology Sheet — Demographics Fields

This sheet lists the approved controlled term set for each demographics field. All future data entry and cleaning should map raw values only to these terms. Codes follow CDISC controlled terminology conventions where a standard exists; internal codes are used otherwise.

4.1 Gender

Controlled Term	Code	Definition
Male	M	Assigned male demographic category
Female	F	Assigned female demographic category
Other	O	Gender identity outside binary categories
Not Reported	NR	Value missing or refused by subject

4.2 Race

Controlled Term	Code	Definition
White	WHT	CDISC controlled terminology (C41261)
Black or African American	BLK	CDISC controlled terminology (C16352)
Asian	ASN	CDISC controlled terminology (C41260)
Multiple	MULT	Subject reports more than one race category
Unknown	UNK	Race not determined or not reported

4.3 Marital Status

Controlled Term	Code	Definition
Single	SGL	Never married
Married	MAR	Currently married / partnered
Divorced	DIV	Marriage legally dissolved
Widowed	WID	Spouse deceased
Not Reported	NR	Value missing or refused

4.4 Country

Controlled Term	Code	Definition
India	IND	ISO 3166-1 alpha-3 standard
United States	USA	ISO 3166-1 alpha-3 standard
United Kingdom	GBR	ISO 3166-1 alpha-3 standard

4.5 Ethnicity

Controlled Term	Code	Definition
Hispanic or Latino	HISP	CDISC controlled terminology (C41260 subset)
Not Hispanic or Latino	NHISP	CDISC controlled terminology (C41260 subset)
Unknown	UNK	Ethnicity not determined
Not Reported	NR	Value missing or refused by subject

4.6 Employment Status

Controlled Term	Code	Definition
Employed	EMP	Subject is currently working
Unemployed	UEMP	Subject is not currently working
Retired	RET	Subject has formally retired from work
Student	STU	Subject is currently enrolled in education
Not Reported	NR	Value missing or refused by subject

5. AI-Supported Mapping Suggestions with Final Student-Approved Outputs

AI was used to generate first-pass mapping suggestions by clustering similar raw values and matching them to the nearest controlled term. Every AI suggestion was reviewed by the student (CDM specialist role) before being finalized. The summary below shows the acceptance rate per field; full row-level approvals are shown in Section 3 under the 'Status' column.

Field	AI Suggestions	Approved As-Is	Modified by Student	Acceptance Rate
Gender	8	8	0	100%
Race	7	6	1	86%
Country	6	6	0	100%
Marital Status	5	5	0	100%
Date of Birth	3	2	1	67%
Ethnicity	4	4	0	100%
Employment Status	5	4	1	80%

Example of a student-modified AI suggestion:

- Raw value: "Mixed" (Race) — AI suggested "Multiple"; student confirmed this matched CDISC intent but flagged it for guide review since "Mixed" could also indicate missing sub-category detail.
- Raw value: "06-05-90" (Date of Birth) — AI suggested 1990-06-05 assuming MM-DD-YY; student corrected the assumption after cross-checking against the subject's age field and confirmed the final ISO date.

Note: AI-generated suggestions were treated as a first draft only. Final standardized values in this document reflect student review and are ready for guide (Mrs. Nidhi Jha) sign-off.

6. Methodology and Tools Used

The standardization process combined manual CDM review with AI-assisted automation, following the sequence below:

- Data extraction: unique raw values per demographics field were pulled from the messy dummy dataset in Castor EDC (reusing the profiling output from the earlier error-detection task).
- AI clustering: an AI assistant grouped raw values that likely represented the same underlying concept (case differences, typos, abbreviations, punctuation, local-language names).
- Terminology matching: each cluster was matched to the nearest controlled term, using CDISC controlled terminology as the primary reference and internal project convention for fields without a CDISC standard (e.g., Employment Status).
- Student review: every AI-suggested mapping was checked line by line; suggestions were approved as-is, corrected, or escalated for guide review.
- Documentation: approved mappings were compiled into the mapping file, the raw-to-standard table, and the controlled terminology sheet in this document.

Tools referenced: Castor EDC (source dataset), CDISC Controlled Terminology (reference standard), AI assistant (first-pass clustering and mapping suggestions), MS Word (final deliverable).

7. Limitations and Assumptions

- This task was performed on a dummy training dataset with a limited number of records; a production dataset would likely surface additional raw-value variants not seen here.
- Date-of-birth standardization assumed a DD/MM/YYYY entry convention where the raw format was ambiguous; this assumption should be confirmed against site-level data entry guidelines before applying to a live study.
- AI-suggested mappings are a first draft only and were not applied to the dataset without student review; any future automation of this step should retain a human-in-the-loop approval gate.
- Controlled terms not covered by CDISC (e.g., Employment Status) used an internally defined term set for this training exercise and would need sponsor/study-level approval in a real study.

8. Abbreviations

Abbreviation	Full Form
CDM	Clinical Data Management
CDISC	Clinical Data Interchange Standards Consortium
EDC	Electronic Data Capture
ISO	International Organization for Standardization

AI	Artificial Intelligence
NR	Not Reported
UNK	Unknown

9. References

- CDISC Controlled Terminology, Clinical Data Interchange Standards Consortium (CDISC).
- Castor EDC — Dummy Clinical Dataset, Demographics Form, CDM Training Task.
- Internal deliverables from this training: Data Profiling Report and Query Rulebook (prior tasks on the same dataset).

10. Summary

- All demographic fields with inconsistent raw entries were mapped to a single controlled standard value.
- A reusable controlled terminology sheet was created for Gender, Race, Ethnicity, Marital Status, Country, and Employment Status.
- AI-assisted clustering reduced manual mapping effort while the student retained final approval authority on every value.
- This mapping file can now be applied to clean the full dataset and can be reused for future standardization tasks in this training.